Chase Mateusiak

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Skills

Experienced bioinformatician with fluency in R and Python. I am comfortable in a linux environment and have years of experience on clusters using both SLURM and SGE. I am responsible for providing two labs with AWS access, and have provided Seqera Platform to them. Seqera platform is a commercial product that simplifies and streamlines running reproducible Nextflow based workflows, and procuring computational resources for collaborative analysis of the results, in a cost effective manner.

I interact with multiple open source development projects and routinely train students on the effective use of git and github.

I am equally comfortable automating standardized data processing workflows and conducting analysis using both supervised and unsupervised machine learning methods.

Work

Bioinformatics Scientist

Washington University, St. Louis, *2020–2026*

Primary responsibilities

- Implement established, recently published, and novel analysis algorithms on bulk RNA-seq, various transcription factor binding modalities (Calling Cards, ChEC-seq, ChIP-seq) and sequence analysis in both single data type and multi-omic modes
- Responsible for data processing and QC of sequencing experiment data for multiple labs, including a large $80 million multi-site collaboration
- Provide computational support to collaborators across organisms (yeast, mouse, human)

Notable achievements

- Developed a novel method of inferring transcription factor interaction at promoters, integrating both transcription factor binding and overexpression and knockout (Mateusiak et al, currently out for review 2026)
- Collected, harmonized, databased all 11 of the major yeast transcription factor binding and perturbation response datasets (Mateusiak et al, expected publication 2026). Notably, the data is hosted using a data lake architecture on huggingface, a machine learning platform that provides free hosting. I developed both a programmatic library and shiny app that provides researchers easy access to this rich data
- Developed, published and released on nf-core a Nextflow workflow for processing Calling Cards data in both yeast and mammals
- Implemented and manage the cloud architecture for two labs, including providing access to Seqera Platform
- Developed independent collaborations resulting in two publications
- Provided computational support to a clinical group by implementing and evaluating the performance of four of the leading circadian rhythm inference algorithms
- Developed the Python curriculum implementation for Algorithms for Computational Biology, a 500-level course in the Washington University CS department
- Trained and directly supervised 5 undergraduate and master's students in machine learning based methods of multi-omics data analysis

Research Assistant

University of California, San Francisco, *2019*

- Processed RNA-seq data from human skin cells and tick
- Compiled transcriptome annotations for *Ixodes scapularis*
- Processed data from a deep mutational scan screen in *E. coli*

Teacher

San Francisco Unified School District, *2012–2019*

- Responsible for the academic progress of 115 students per year in English and math

Civil Affairs Sergeant

US Army Reserve, *2011–2019*

- Trained in developing local relationships, identifying achievable development projects, and coordinating allied forces activities with local government and non-profit organizations

Teacher

US Peace Corps, Turkmenistan, *2008–2010*

- Developed and implemented an interactive English as a Foreign Language curriculum that improved speaking and vocabulary skills across eight classrooms for over 250 students

Education

- M.S. Computer Science — Washington University, St. Louis, *2022*
- B.A. English Literature and Language — University of Michigan, Ann Arbor, *2008*

Publications

Mateusiak, C., Jia, E., Plaggenberg, J. N., Erdenebaatar, Z., Wang, Y., Shively, C., Liao, G., Mitra, R. D., & Brent, M. R. (2026). Ubiquitous functional synergy partially explains why most transcription factor binding is non-functional. *bioRxiv* (preprint). https://doi.org/10.64898/2026.01.19.700460

Karagiannis, T. T., Mateusiak, C., Acharya, S., Jung, W. J., Liao, S., et al. (2026). Whole blood transcriptional signatures of age and survival identified in long life family and integrative longevity omics studies. *GeroScience*. https://doi.org/10.1007/s11357-025-02090-x

Abid, D., Brown, H., Mateusiak, C., Doering, T. L., & Brent, M. R. (2025). Mapping the transcriptional regulatory network of a fungal pathogen by exploiting transcription factor perturbation. *mBio*. https://doi.org/10.1128/mbio.02797-25

Kang, Y. S., Jung, J., Brown, H. L., Mateusiak, C., Doering, T. L., & Brent, M. R. (2025). Leveraging a new data resource to define the response of *Cryptococcus neoformans* to environmental signals. *Genetics*, 229(1), 1–29. https://doi.org/10.1093/genetics/iyae178

Yen, A., Mateusiak, C., Sarafinovska, S., Gachechiladze, M. A., Guo, J., Chen, X., Moudgil, A., Cammack, A. J., Hoisington-Lopez, J., Crosby, M., Brent, M. R., Mitra, R. D., & Dougherty, J. D. (2023). Calling Cards: a customizable platform to longitudinally record protein-DNA interactions over time in cells and tissues. *Current Protocols*, 3(9), e883. https://doi.org/10.1002/cpz1.883

Ring, K., Couper, L. I., Sapiro, A. L., Yarza, F., Yang, X. F., Clay, K., Mateusiak, C., Chou, S., & Swei, A. (2022). Host blood meal identity modifies vector gene expression and competency. *Molecular Ecology*, 31(9), 2698–2711. https://doi.org/10.1111/mec.16413

Radkov, A., Sapiro, A. L., Flores, S., Henderson, C., Saunders, H., Kim, R., Massa, S., Thompson, S., Mateusiak, C., Biboy, J., Zhao, Z., Starita, L. M., Hatleberg, W. L., Vollmer, W., Russell, A. B., Simorre, J. P., Anthony-Cahill, S., Brzovic, P., Hayes, B., & Chou, S. (2022). Antibacterial potency of type VI amidase effector toxins is dependent on substrate topology and cellular context. *eLife*, 11, e79796. https://doi.org/10.7554/eLife.79796